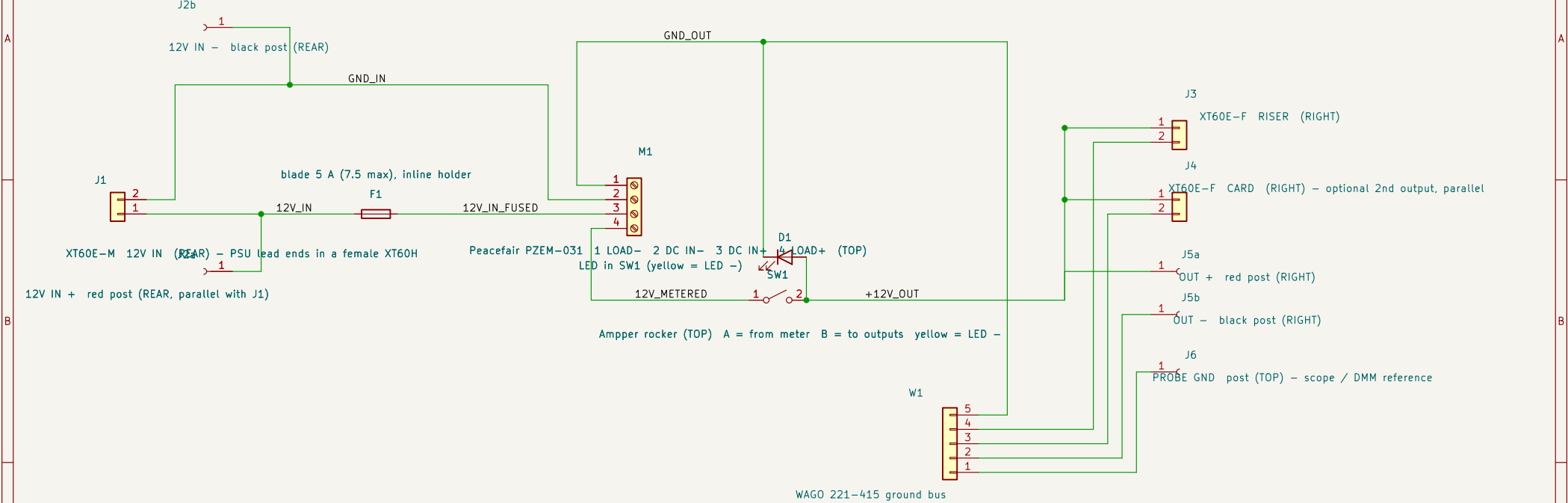


PCIe GPU test rig – power box wiring

Rev C 2026-09-28. XT60 panel connectors replace the pigtail; meter ahead of the switch; WAGO ground bus. PZEM-031 terminals top->bottom 1..4.



Path: 12V IN -> F1 -> M1 (3 in, 4 out) -> SW1 -> outputs. The meter reads whenever the supply is on; the switch only gates the outputs.

M1 PZEM-031: shunt is internal between 2 (DC IN-) and 1 (LOAD-). Input negative goes to 2, NOT to the bus - on the bus it would bypass the shunt and the meter would read 0 A.

W1 WAGO 221-415, five ports: 5 = LOAD-, 4 = RISER -, 3 = CARD -, 2 = OUT - post, 1 = PROBE GND. The rocker's yellow LED - rides on the OUT - post's terminal.

SW1: A (power) from M1.4, B (load) to the outputs, yellow to ground. NEVER swap B and yellow - that is a dead short the moment it is switched on. Beep the pins first.

Ratings: OWON at 12.0 V, OVP ~12.6 V, current limit 5 A (CC for short hunting). F1 5 A blade (7.5 max). Switch 20 A, XT60 30 A, 14 AWG silicone inside. J4 is optional - fit one or two outputs.

Panels: REAR = J1, J2a, J2b. TOP deck = M1, SW1, J6 (+ riser dock). RIGHT wall = J3, J4, J5a, J5b. F1 and W1 are inline inside.

pcie-rig-enclosure/wiring - rev C 2026-09-28, edit in KiCad
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Title: PCIe GPU test rig – power box wiring

Size: A4 Date: 2026-09-28

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